

BrightLearn data Analytics Coding Practical Exercise 1

--1. SELECT Statement

--Q1.

```
SELECT *  
FROM `workspace`.`default`.`retail_sales_dataset` limit 5;
```

Output:

	1 ² ₃ Transaction ID	📅 Date	A ^B _C Customer ID	A ^B _C Gender	1 ² ₃ Age	A ^B _C Product Category	1 ² ₃ Quantity
1	1	2023-11-24	CUST001	Male	34	Beauty	3
2	2	2023-02-27	CUST002	Female	26	Clothing	2
3	3	2023-01-13	CUST003	Male	50	Electronics	1
4	4	2023-05-21	CUST004	Male	37	Clothing	1
5	5	2023-05-06	CUST005	Male	30	Beauty	2

1 ² ₃ Price per Unit	1 ² ₃ Total Amount
50	150
500	1000
30	30
500	500
50	100

--Q2.

```
SELECT `Transaction ID`,  
       Date,  
       `Customer ID`  
FROM `workspace`.`default`.`retail_sales_dataset`;
```

Output:

	1 ² ₃ Transaction ID	📅 Date	A ^B _C Customer ID
1	1	2023-11-24	CUST001
2	2	2023-02-27	CUST002
3	3	2023-01-13	CUST003
4	4	2023-05-21	CUST004
5	5	2023-05-06	CUST005

--- 2. SELECT DISTINCT

-- Q3

```
SELECT DISTINCT `Product Category`  
FROM `workspace`.`default`.`retail_sales_dataset`;
```

Output:

	A_C^B Product Category
1	Beauty
2	Clothing
3	Electronics

-- Q4

```
SELECT DISTINCT Gender  
FROM `workspace`.`default`.`retail_sales_dataset`;
```

Output:

	A_C^B Gender
1	Male
2	Female

---3. WHERE Clause

-- Q5

```
SELECT *  
FROM `workspace`.`default`.`retail_sales_dataset`  
WHERE Age > 40;
```

Output:

	I_3^2 Transaction ID	 Date	A_C^B Customer ID	A_C^B Gender	I_3^2 Age	A_C^B Product Category	I_3^2 Quantity
1	3	2023-01-13	CUST003	Male	50	Electronics	1
2	6	2023-04-25	CUST006	Female	45	Beauty	1
3	7	2023-03-13	CUST007	Male	46	Clothing	2
4	9	2023-12-13	CUST009	Male	63	Electronics	2
5	10	2023-10-07	CUST010	Female	52	Clothing	4

¹ ₃ Price per Unit	¹ ₃ Total Amount
30	30
30	30
25	50
300	600
50	200

-- Q6

```
SELECT *
FROM `workspace`.`default`.`retail_sales_dataset`
WHERE `Price per Unit` BETWEEN 100 AND 500;
```

Output:

	¹ ₃ Transaction ID	 Date	^A _C Customer ID	^A _C Gender	¹ ₃ Age	^A _C Product Category	¹ ₃ Quantity
1	2	2023-02-27	CUST002	Female	26	Clothing	2
2	4	2023-05-21	CUST004	Male	37	Clothing	1
3	9	2023-12-13	CUST009	Male	63	Electronics	2
4	13	2023-08-05	CUST013	Male	22	Electronics	3
5	15	2023-01-16	CUST015	Female	42	Electronics	4

¹ ₃ Price per Unit	¹ ₃ Total Amount
500	1000
500	500
300	600
500	1500
500	2000

-- Q7

```
SELECT *
FROM `workspace`.`default`.`retail_sales_dataset`
WHERE `Product Category` IN ('Beauty', 'Electronics');
```

Output:

	¹ ₃ Transaction ID	 Date	^A _C Customer ID	^A _C Gender	¹ ₃ Age	^A _C Product Category
1	1	2023-11-24	CUST001	Male	34	Beauty
2	3	2023-01-13	CUST003	Male	50	Electronics
3	5	2023-05-06	CUST005	Male	30	Beauty
4	6	2023-04-25	CUST006	Female	45	Beauty
5	8	2023-02-22	CUST008	Male	30	Electronics

1^2_3 Quantity	1^2_3 Price per Unit	1^2_3 Total Amount
3	50	150
1	30	30
2	50	100
1	30	30
4	25	100

-- Q8

```
SELECT *
FROM `workspace`.`default`.`retail_sales_dataset`
WHERE `Product Category` != 'Clothing';
```

Output:

	1^2_3 Transaction ID	 Date	A^B_C Customer ID	A^B_C Gender	1^2_3 Age	A^B_C Product Category
1	1	2023-11-24	CUST001	Male	34	Beauty
2	3	2023-01-13	CUST003	Male	50	Electronics
3	5	2023-05-06	CUST005	Male	30	Beauty
4	6	2023-04-25	CUST006	Female	45	Beauty
5	8	2023-02-22	CUST008	Male	30	Electronics

1^2_3 Quantity	1^2_3 Price per Unit	1^2_3 Total Amount
3	50	150
1	30	30
2	50	100
1	30	30
4	25	100

-- Q9

```
SELECT *
FROM `workspace`.`default`.`retail_sales_dataset`
WHERE Quantity >= 3;
```

Output:

	1^2_3 Transaction ID	 Date	A^B_C Customer ID	A^B_C Gender	1^2_3 Age	A^B_C Product Category
1	1	2023-11-24	CUST001	Male	34	Beauty
2	8	2023-02-22	CUST008	Male	30	Electronics
3	10	2023-10-07	CUST010	Female	52	Clothing
4	12	2023-10-30	CUST012	Male	35	Beauty
5	13	2023-08-05	CUST013	Male	22	Electronics

1 ² ₃ Quantity	1 ² ₃ Price per Unit	1 ² ₃ Total Amount
3	50	150
4	25	100
4	50	200
3	25	75
3	500	1500

--- 4. Aggregate Functions

-- Q10

```
SELECT COUNT(*) AS Total_Transactions
FROM `workspace`.`default`.`retail_sales_dataset`;
```

Output:

	1 ² ₃ Total_Transactions
1	1000

-- Q11

```
SELECT AVG(Age) AS Average_Age
FROM `workspace`.`default`.`retail_sales_dataset`;
```

Output:

	1.2 Average_Age
1	41.392

-- Q12

```
SELECT SUM(Quantity) AS Total_Quantity
FROM `workspace`.`default`.`retail_sales_dataset`;
```

Output:

	1 ² ₃ Total_Quantity
1	2514

-- Q13

```
SELECT MAX(`Total Amount`) AS Max_Total_Amount
FROM `workspace`.`default`.`retail_sales_dataset`;
```

Output:

	1 ² ₃ Max_Total_Amount
1	2000

-- Q14

```
SELECT MIN(`Price per Unit`) AS Min_Price_per_Unit
FROM `workspace`.`default`.`retail_sales_dataset`;
```

Output:

	1 ² ₃ Min_Price_per_Unit
1	25

---5. GROUP BY

-- Q15

```
SELECT `Product Category`, COUNT(*) AS Transaction_Count
FROM `workspace`.`default`.`retail_sales_dataset`
GROUP BY `Product Category`;
```

Output:

	A ^B _C Product Category	1 ² ₃ Transaction_Count
1	Beauty	307
2	Clothing	351
3	Electronics	342

-- Q16

```
SELECT Gender, SUM(`Total Amount`) AS Total_Revenue
FROM `workspace`.`default`.`retail_sales_dataset`
GROUP BY Gender;
```

Output:

	A ^B _C Gender	1 ² ₃ Total_Revenue
1	Male	223160
2	Female	232840

-- Q17

```
SELECT `Product Category`, AVG(`Price per Unit`) AS Average_Price
FROM `workspace`.`default`.`retail_sales_dataset`
GROUP BY `Product Category`;
```

Output:

	^A _C Product Category	1.2 Average_Price
1	Beauty	184.05537459283389
2	Clothing	174.28774928774928
3	Electronics	181.90058479532163

----6. HAVING Clause

-- Q18

```
SELECT `Product Category`, SUM(`Total Amount`) AS Total_Revenue
FROM `workspace`.`default`.`retail_sales_dataset`
GROUP BY `Product Category`
HAVING SUM(`Total Amount`) > 10000;
```

Output:

	^A _C Product Category	¹ ₂ Total_Revenue
1	Beauty	143515
2	Clothing	155580
3	Electronics	156905

-- Q19

```
SELECT `Product Category`, AVG(Quantity) AS Average_Quantity
FROM `workspace`.`default`.`retail_sales_dataset`
GROUP BY `Product Category`
HAVING AVG(Quantity) > 2;
```

Output:

	^A _C Product Category	1.2 Average_Quantity
1	Beauty	2.511400651465798
2	Clothing	2.547008547008547
3	Electronics	2.482456140350877

```

---7. CASE Statement
-- Q20
SELECT
    `Transaction ID`,
    `Total Amount`,
    CASE
        WHEN `Total Amount` > 1000 THEN 'High'
        ELSE 'Low'
    END AS Spending_Level
FROM `workspace`.`default`.`retail_sales_dataset`;

```

Output:

	¹ ₃ Transaction ID	¹ ₃ Total Amount	^A _C Spending_Level
1	1	150	Low
2	2	1000	Low
3	3	30	Low
4	4	500	Low
5	5	100	Low

```

-- Q21
SELECT
    `Customer ID`,
    Age,
    CASE
        WHEN Age < 30 THEN 'Youth'
        WHEN Age BETWEEN 30 AND 59 THEN 'Adult'
        ELSE 'Senior'
    END AS Age_Group
FROM `workspace`.`default`.`retail_sales_dataset`;

```

Output:

	^A _C Customer ID	¹ ₃ Age	^A _C Age_Group
1	CUST001	34	Adult
2	CUST002	26	Youth
3	CUST003	50	Adult
4	CUST004	37	Adult
5	CUST005	30	Adult